

Supplementary Information

A Provenance-Aware Knowledge Graph for Role-Adaptive Access to Traditional Chinese Food Therapy Records

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Purpose and scope

This Supplementary Information documents the recoverable evidence underlying Revision 12. It distinguishes expert reference annotations, frozen model outputs, sampled rule validation, OCR audits and participant-level analyses. Candidate records, procedural flags and application views have different meanings and are not combined into one inferred-fact total. Historical source assertions are not modern evidence of efficacy, safety, diagnosis or treatment.

The submission supplement comprises this merged PDF and two separately numbered Supplementary Data files. A larger data-and-code release candidate exists locally but is not represented here as a completed public deposit. No repository DOI or accession has been issued. Copyrighted page scans and complete source books are excluded unless redistribution rights permit release.

S1. Recoverable archive and expert-reference scope

The corpus-processing inventory contains 52 full-extraction files. The source crosswalk has 53 source keys because one annotation-frame source did not belong to the extraction corpus. Forty annotation-frame source identifiers were resolved and five filenames remained unresolved. These numbers describe the recoverable archival structure and do not constitute a probability sample of all traditional Chinese food-therapy texts.

Two experts annotated independently and a third expert adjudicated disagreements. The archived adjudicated reference contains 763 passages and 18,052 entity rows. It remains separate from the automatically generated production graph. The final release audit identified character-offset drift between the archived entity rows and the current passage version. The counts below therefore document the recovered annotation inventory, not a release-validated strict-span benchmark.

Table 1: Recoverable corpus and reference-set accounting.

Item	Records	Source keys	Interpretation
Full-extraction corpus	52 files	52	Purposive archive included in extraction
Source crosswalk	53 keys	53	Includes one annotation-frame-only source
Resolved annotation-frame sources	40 identifiers	40	Recoverable links to named sources
Unresolved filenames	5 files	–	No defensible source assignment available
Archived adjudicated reference	763 passages	–	Independent annotation followed by adjudication
Archived entity rows	18,052	–	Mention inventory; offsets require realignment
Canonical graph vocabulary	82,522 entities	–	Automatically produced canonical records
Source-derived graph	33,805 triples	–	Model-verified records, not clinical facts

S2. Entity-export integrity and relation evaluation

The final NER release audit compared every declared entity string with the current passage version. Only 4,858 of 18,052 rows aligned at their declared offsets. Another 12,385 strings occurred elsewhere in the passage, and 809 were absent from the current passage version. This pattern indicates passage-version or character-offset drift. Automatic nearest-string movement would be unsafe when strings repeat or no exact string remains. The archived strict-span scores of 0.066, 0.090, 0.252 and 0.256 are retained only as historical analysis outputs. They must be recomputed after one passage version is frozen and the realigned spans receive expert confirmation.

Table 2: Integrity audit of the archived entity export.

Audit status	Entity rows	Release interpretation
Declared span exactly matches entity text	4,858	Exact alignment in the current passage version
Entity text occurs elsewhere in the passage	12,385	Realignment and ambiguity review required
Entity text absent from current passage	809	Passage version or text requires expert review
Total	18,052	Strict-span release validation remains incomplete

The confirmatory relation set is not affected by this offset issue. It contains 1,162 type-compatible pairs from 65 previously held-out passages. Two experts reviewed every pair blind to system predictions. They agreed on 1,065 labels, and a third expert adjudicated the remaining 97. Final labels comprised 445 present, 575 absent, 97 insufficient-context and 45 entity-or-type-error cases. The primary analysis treated all non-present categories as negative. Confidence intervals were obtained by resampling passages.

Table 3: Human-revalidated relation classification on the frozen confirmatory set.

Method	Precision	Recall	F1	F1 lower	F1 upper	Predicted
Typed co-occurrence	0.383	1.000	0.554	0.492	0.610	1,162
Distance-trigger heuristic	0.470	0.591	0.523	0.447	0.594	560
Fixed character n-gram comparator	0.366	0.551	0.439	0.373	0.508	670
DeepSeek supported-or-plausible	0.665	0.317	0.429	0.317	0.522	212
DeepSeek supported-only	0.694	0.301	0.420	0.298	0.518	193
Local symbolic-text comparator	0.361	0.499	0.419	0.338	0.497	615
CMFKG frozen hybrid	0.361	0.499	0.419	0.336	0.500	615

CMFKG did not outperform the fixed comparator, distance heuristic or typed co-occurrence method on relation F1. The manuscript therefore makes no relation-extraction superiority claim.

S3. OCR audit and downstream propagation sensitivity

Sixty pages were sampled for transcription. Fifty-one had adjudicated manual transcriptions and 49 distinct source keys. Forty-two pages, representing 40 source keys, crosswalked to the 52-source extraction corpus and form the primary corpus-scoped raw OCR analysis. Nine adjudicated pages lay outside that crosswalk. Seventeen corpus-linked pages had raw OCR, cleaned OCR and adjudicated transcription; cleanup results are restricted to this paired subset.

Table 4: OCR audit estimates and reporting scope.

Analysis	Pages	Mean CER	95% CI	Scope
Primary raw OCR audit	42	14.88%	10.25–19.98%	Corpus-crosswalked pages; source-key cluster bootstrap
Broader technical raw audit	51	14.91%	11.34–18.94%	All adjudicated pages; retained for provenance
Cleaned OCR	17	1.14%	0.22–2.59%	Raw/cleaned paired corpus subset only
Cleaned minus raw CER	17	-26.39 points	-35.22 to -18.12	Paired subset only

The primary 42-page raw CER had a median of 8.09%, an interquartile range of 5.89–15.78% and a range of 2.10–63.34%. The cleaned value must not be generalised to all 51 pages. The archived workbook shows that the adjudicator selected the second transcription as the final text on all 51 pages. This selection pattern is disclosed for audit transparency.

Downstream propagation compares frozen predictions on raw or cleaned OCR with predictions on adjudicated transcription. It measures output stability, not human-annotated accuracy or automatic OCR-error detection. Across the 17 paired pages, entity stability F1 improved from 0.846 to 0.980 for SikuRoBERTa-CRF and from 0.808 to 0.988 for Chinese-RoBERTa-CRF. No detector precision or recall is claimed because no OCR-error detector was evaluated.

S4. Operational-rule evidence

The 13 operational specifications were designed from the schema, historical relation paths and interface requirements. They are not clinical guidelines and did not receive prospective expert approval. Their outputs remain candidates, audit flags or display policies until separately reviewed.

Three corrected B-class candidate populations were sampled independently with 100 items per rule. Two experts reviewed the samples before adjudication. Supported proportions were 0% for B1, 42% for B2 and 8% for B3. Uncertain labels were retained rather than collapsed into support.

Table 5: Human review of three B-class rule samples.

Rule	Sample	Supported	Not supported	Uncertain	Support	95% Wilson CI
B1	100	0	99	1	0%	0.0–3.7%
B2	100	42	10	48	42%	32.8–51.8%
B3	100	8	59	33	8%	4.1–15.0%

The archived A-class flow contains 4,747 raw candidates, of which 2,234 passed vocabulary filtering. The legacy combined B-class file contains 49,099 exploratory candidates without separable per-rule provenance. C-class processing produced 341 unadjudicated procedural flags, including 48 C1, 292 C2 and one C3. D-class outputs are overlapping views of asserted graph records, with 4,417 visible records for general users and 33,805 each for TCM professionals and researchers. These values are not additive.

S5. Participant-level evaluation results

The study recruited 106 participants. Six withdrew or met recorded exclusion criteria before randomisation. The final participant-level analysis included 100 participants: 34 general users, 33 TCM professionals and 33 medical researchers. Eight interrupted task submissions had no scorable response and were excluded only at task level, leaving 2,392 scored task responses. Every participant used both systems in a role-stratified, counterbalanced design. The Ethics Committee of Beijing Institute of Technology approved the study under BIT-EC-HM-2026019. All participants gave written informed consent.

The participant was the independent unit. Two primary endpoints were tested with paired t-tests, 95% confidence intervals and paired Cohen d_z . Holm adjustment covered both endpoints. Confidence intervals for d_z used 10,000 participant-level bootstrap samples with seed 20260731. Role differences were assessed by one-way analysis of variance on participant-level paired differences.

Table 6: Primary paired participant-level outcomes.

Outcome	CMFKG	Traditional	Difference	95% CI	Paired d_z (95% CI)
Accuracy, 0–100	79.18	69.68	9.49	8.53–10.46	1.951 (1.677–2.301)
Weighted NASA-TLX	32.42	42.63	-10.22	-11.43 to -9.01	-1.673 (-2.020 to -1.419)

The Holm-adjusted P values were 2.15×10^{-35} for accuracy and 1.28×10^{-30} for workload. The role-by-system difference was not detected for accuracy ($F = 0.886$, $P = 0.416$), whereas paired workload differences varied across roles ($F = 5.041$, $P = 0.0083$). These outcomes apply to the complete CMFKG system and do not isolate a causal effect of the type-derived L1–L6 presentation field.

S6. Supplementary Data inventory and release status

Supplementary Data 1 is a multi-sheet workbook containing the corpus source inventory, 800-passage disposition, NER offset-integrity audit, human-revalidated relation labels and metrics, rule-validation items and summaries, OCR page and summary results, participant-level aggregate results, and figure and table source data.

Supplementary Data 2 is a ZIP archive containing three machine-readable resources. The consolidated entity-annotation guideline records the archived expert workflow, 13 entity classes, boundary rules, adjudication logic and the current offset-integrity limitation. The CMFKG ontology extension formalises the local classes, relations and provenance properties in Turtle syntax without claiming that the computational graph was stored natively as CIDOC CRM RDF. The operational-rule inventory records the antecedents, conditions, implementation links, outputs, validation status and known limitations for the 13 specifications.

The larger local release candidate also contains frozen predictions, analysis scripts and environment records. Credential and direct-identifier checks have been completed. Public deposit remains blocked until the NER offset realignment and expert confirmation are complete. No DOI, accession or completed public archival deposit is claimed in this supplement.